OS Skill-23

To Execute Large Pipelines, Launch Multiple Jobs, Measure Stability, Monitor Resources, Detect Failures, Analyze Performance.

To Document Architecture, Create Flow Diagrams, Explain Execution Paths, Describe Syscalls, Record Design Decisions, Review Documentation.

```
lahari@DESKTOP-AIC33F7:~$ mkdir pipeline_lab
cd pipeline_lab
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ seq 1 100000 > numbers.txt
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ wc -l numbers.txt
100000 numbers.txt
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ cat numbers.txt | grep '99' | sort -n | head -20
99
199
299
399
499
599
699
799
899
990
991
992
993
994
995
996
997
998
999
1099
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ cat numbers.txt | grep '99' | wc -l
3691
```

```
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ sort -n numbers.txt > sorted.txt &
grep '99' numbers.txt > matches.txt &
wc -l numbers.txt > count.txt &
[1] 889
[2] 890
[3] 891
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ jobs -l
[1]    889 Done                    sort -n numbers.txt > sorted.txt
[2]-   890 Done                    grep --color=auto '99' numbers.txt > matches.txt
[3]+   891 Done                    wc -l numbers.txt > count.txt
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ ps aux | head
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1  21864 13216 ?        Ss   05:10   0:00 /sbin/init
root           2  0.0  0.0   3180  2208 hvc0     Sl+  05:10   0:00 /init
root           6  0.0  0.0   3196  2128 hvc0     Sl+  05:10   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10
--log-truncate
root          52  0.0  0.2  50448 16364 ?        S<s  05:10   0:00 /usr/lib/systemd/systemd-journald
root          99  0.0  0.0  25020  6400 ?        Ss   05:10   0:00 /usr/lib/systemd/systemd-udevd
systemd+     114  0.0  0.1  21344 13288 ?        Ss   05:10   0:00 /usr/lib/systemd/systemd-resolved
systemd+     115  0.0  0.1  91036  8048 ?        Ssl  05:10   0:00 /usr/lib/systemd/systemd-timesyncd
root         163  0.0  0.0   4244  2748 ?        Ss   05:10   0:00 /usr/sbin/cron -f -P
message+     164  0.0  0.0   9676  5544 ?        Ss   05:10   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --sy
stemd-activation --syslog-only
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ top
top - 05:22:50 up 22 min,  1 user,  load average: 0.04, 0.01, 0.00
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7643.6 total,   6958.5 free,    504.9 used,    331.6 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7138.7 avail Mem
```

```
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ top
top - 05:22:50 up 22 min,  1 user,  load average: 0.04, 0.01, 0.00
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7643.6 total,   6958.5 free,    504.9 used,    331.6 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7138.7 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
     99 root      20   0   25020   6400   5212 S   0.3   0.1   0:00.23 systemd-udevd
      1 root      20   0   21864  13216   9864 S   0.0   0.2   0:00.58 systemd
      2 root      20   0    3180   2208   2072 S   0.0   0.0   0:00.00 init-systemd(Ub
      6 root      20   0    3196   2128   2008 S   0.0   0.0   0:00.00 init
     52 root      19  -1   50448  16364  15208 S   0.0   0.2   0:00.21 systemd-journal
    114 systemd+  20   0   21344  13288  11168 S   0.0   0.2   0:00.09 systemd-resolve
    115 systemd+  20   0   91036   8048   7084 S   0.0   0.1   0:00.07 systemd-timesyn
    163 root      20   0    4244   2748   2496 S   0.0   0.0   0:00.01 cron
    164 message+  20   0    9676   5544   4808 S   0.0   0.1   0:00.08 dbus-daemon
    175 root      20   0   17976   8884   7840 S   0.0   0.1   0:00.06 systemd-logind
    184 syslog    20   0  222516   5672   4644 S   0.0   0.1   0:00.05 rsyslogd
    193 root      20   0    3124   2040   1900 S   0.0   0.0   0:00.00 agetty
    199 root      20   0  107036  23304  13844 S   0.0   0.3   0:00.09 unattended-upgr
    290 root      20   0    6704   4732   3948 S   0.0   0.1   0:00.01 login
    338 lahari    20   0   20124  11452   9520 S   0.0   0.1   0:00.14 systemd
    339 lahari    20   0   21168   3584   1860 S   0.0   0.0   0:00.00 (sd-pam)
    361 lahari    20   0    6080   5376   3656 S   0.0   0.1   0:00.03 bash
    653 polkitd   20   0  308172   8188   7296 S   0.0   0.1   0:00.09 polkitd
    847 root      20   0    3184   1120    980 S   0.0   0.0   0:00.00 SessionLeader
    848 root      20   0    3200   1256   1108 S   0.0   0.0   0:00.01 Relay(853)
    853 lahari    20   0    6728   6008   3704 S   0.0   0.1   0:00.04 bash
    894 lahari    20   0    9284   5728   3556 R   0.0   0.1   0:00.00 top
    895 root      20   0   25024   3496   2276 S   0.0   0.0   0:00.00 (udev-worker)
    896 root      20   0   25024   3496   2276 S   0.0   0.0   0:00.00 (udev-worker)
top - 05:23:32 up 23 min,  1 user,  load average: 0.02, 0.01, 0.00
```

```
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ free -h
               total        used        free      shared  buff/cache   available
Mem:           7.5Gi       500Mi       6.8Gi       3.5Mi       331Mi       7.0Gi
Swap:          2.0Gi          0B       2.0Gi
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ df -h .
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdd       1007G  2.7G  953G   1% /
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ time sort -n numbers.txt > sorted.txt

real    0m0.047s
user    0m0.043s
sys     0m0.005s
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ sort missing.txt
echo $?
sort: cannot read: missing.txt: No such file or directory
2
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ set -o pipefail
cat missing.txt | sort | wc -l
echo $?
cat: missing.txt: No such file or directory
0
1
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ wait
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ ls -lh
cat count.txt
wc -l sorted.txt
wc -l matches.txt
total 1.2M
-rw-r--r-- 1 lahari lahari   19 Oct 13 05:22 count.txt
-rw-r--r-- 1 lahari lahari  22K Oct 13 05:22 matches.txt
-rw-r--r-- 1 lahari lahari 576K Oct 13 05:21 numbers.txt
-rw-r--r-- 1 lahari lahari 576K Oct 13 05:23 sorted.txt
```

```
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ wait
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ ls -lh
cat count.txt
wc -l sorted.txt
wc -l matches.txt
total 1.2M
-rw-r--r-- 1 lahari lahari   19 Oct 13 05:22 count.txt
-rw-r--r-- 1 lahari lahari  22K Oct 13 05:22 matches.txt
-rw-r--r-- 1 lahari lahari 576K Oct 13 05:21 numbers.txt
-rw-r--r-- 1 lahari lahari 576K Oct 13 05:23 sorted.txt
100000 numbers.txt
100000 sorted.txt
3691 matches.txt
```

```
3891 matches.txt
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ {
    echo "Pipeline Performance Report"
    date
    time sort -n numbers.txt > sorted.txt
    echo "Sorted output line count:"
    wc -l sorted.txt
    echo "Memory information:"
    free -h
} 2>&1 | tee performance_report.txt
Pipeline Performance Report
Tue Oct 13 05:24:23 UTC 2026

real    0m0.038s
user    0m0.021s
sys     0m0.020s
Sorted output line count:
100000 sorted.txt
Memory information:
               total        used        free      shared  buff/cache   available
Mem:           7.5Gi       505Mi       6.8Gi       3.5Mi       331Mi       7.0Gi
Swap:          2.0Gi          0B       2.0Gi
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ cat performance_report.txt
Pipeline Performance Report
Tue Oct 13 05:24:23 UTC 2026

real    0m0.038s
user    0m0.021s
sys     0m0.020s
Sorted output line count:
100000 sorted.txt
Memory information:
               total        used        free      shared  buff/cache   available
Mem:           7.5Gi       505Mi       6.8Gi       3.5Mi       331Mi       7.0Gi
Swap:          2.0Gi          0B       2.0Gi
lahari@DESKTOP-AIC33F7:~/pipeline_lab$ mkdir architecture_lab
cd architecture_lab
```

nano copy.c

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
   int source, destination;
   char buffer[1024];
   ssize_t bytes;

   source = open("input.txt", O_RDONLY);
   if (source == -1) {
      perror("Source open failed");
      return 1;
   }
```

```
    destination = open("output.txt",
                  O_WRONLY | O_CREAT | O_TRUNC,
                  0644);

    if (destination == -1) {
        perror("Destination open failed");
        close(source);
        return 1;
    }

    while ((bytes = read(source, buffer, sizeof(buffer))) > 0) {
        ssize_t written = 0;

        while (written < bytes) {
            ssize_t n = write(destination,
                        buffer + written,
                        bytes - written);

            if (n <= 0) {
                perror("Write failed");
                close(source);
                close(destination);
                return 1;
            }

            written += n;
        }
    }

    if (bytes == -1) {
        perror("Read failed");
        close(source);
        close(destination);
        return 1;
    }

    close(source);
    close(destination);

    printf("File copied successfully\n");
    return 0;
}
```

```
cd architecture_lab
lahari@DESKTOP-AIC33F7:~/pipeline_lab/architecture_lab$ nano copy.c
lahari@DESKTOP-AIC33F7:~/pipeline_lab/architecture_lab$ echo "Welcome to Ubuntu Linux" > input.txt
lahari@DESKTOP-AIC33F7:~/pipeline_lab/architecture_lab$ gcc -Wall -Wextra copy.c -o copy
lahari@DESKTOP-AIC33F7:~/pipeline_lab/architecture_lab$ ./copy
File copied successfully
lahari@DESKTOP-AIC33F7:~/pipeline_lab/architecture_lab$ cat output.txt
Welcome to Ubuntu Linux
lahari@DESKTOP-AIC33F7:~/pipeline_lab/architecture_lab$ sudo apt update
sudo apt install strace -y
[sudo] password for lahari:
 strace -y
Sorry, try again.
[sudo] password for lahari:
LSorry, try again.
[sudo] password for lahari:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
124 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
strace is already the newest version (6.8-0ubuntu2).
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libwayland-server0 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 124 not upgraded.
lahari@DESKTOP-AIC33F7:~/pipeline_lab/architecture_lab$ strace -o syscall_log.txt ./copy
File copied successfully
lahari@DESKTOP-AIC33F7:~/pipeline_lab/architecture_lab$ grep -E 'openat|read|write|close' syscall_log.txt
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
close(3)                                = 0
close(1)                                = 0
write(1, "File copied successfully\n", 25) = 25
lahari@DESKTOP-AIC33F7:~/pipeline_lab/architecture_lab$ nano architecture.md
lahari@DESKTOP-AIC33F7:~/pipeline_lab/architecture_lab$ mv input.txt backup.txt
lahari@DESKTOP-AIC33F7:~/pipeline_lab/architecture_lab$ ./copy
Source open failed: No such file or directory
lahari@DESKTOP-AIC33F7:~/pipeline_lab/architecture_lab$ mv backup.txt input.txt
lahari@DESKTOP-AIC33F7:~/pipeline_lab/architecture_lab$ ls -l
cat architecture.md
total 36
-rw-r--r-- 1 lahari lahari  1438 Oct 13 05:58 architecture.md
-rwxr-xr-x 1 lahari lahari 16216 Oct 13 05:56 copy
-rw-r--r-- 1 lahari lahari  1235 Oct 13 05:56 copy.c
-rw-r--r-- 1 lahari lahari    24 Oct 13 05:56 input.txt
-rw-r--r-- 1 lahari lahari    24 Oct 13 05:57 output.txt
-rw-r--r-- 1 lahari lahari  2751 Oct 13 05:57 syscall_log.txt
# File Copy Program – Architecture Documentation

## 1. Objective
Copy the contents of a source file into a destination file using Linux system calls.

## 2. Architecture
- Input: input.txt
- Processing: C program and a 1024-byte buffer
- Output: output.txt
- Operating system: Ubuntu Linux

## 3. Execution Path
1. Open the source file.
2. Open or create the destination file.
3. Read data into a buffer.
4. Write the buffer to the destination.
5. Repeat until the end of the source file.
6. Close both file descriptors.
7. Display the result.

## 4. System Calls
- open(): Opens a file or creates a destination file.
- read(): Reads bytes from the source.
- write(): Writes bytes to the destination.
- close(): Releases a file descriptor.

## 5. Error Handling
The program checks for file-opening, reading, and writing errors. It reports errors using perror() and closes resources when possible.
```

```
## 6. Design Decisions
- A fixed-size buffer limits memory usage.
- A loop supports files larger than the buffer.
- Partial writes are handled by writing until the entire buffer is processed.
- O_TRUNC clears an existing destination file before copying.

## 7. Testing
- Test a valid source file.
- Test a missing source file.
- Verify destination contents.
- Inspect system calls using strace.

## 8. Conclusion
The program demonstrates file I/O, system calls, resource cleanup, error handling, and software architecture documentation.
lahari@DESKTOP-AIC33F7:~/pipeline_lab/architecture_lab$
```